EXPERIMENT 11

Non-Contiguous Memory Allocation: LRU

AIM:

To implement the Least Recently Used (LRU) page replacement algorithm and calculate the total number of page faults.

PROCEDURE:

1: Start the process

2: Declare the size

3: Get the number of pages to be inserted

4: Get the value

5: Declare counter and stack

6: Select the least recently used page by counter value

7: Stack them according the selection.

8: Display the values

9: Stop the process

PROGRAM:

```
#include <stdio.h>

int main()
{
    int frames[10], pages[30], time[10];
    int n, m, i, j, pos, faults = 0, count = 0;
    int found, min;

    printf("Enter number of frames: ");
    scanf("%d", &m);

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter reference string: ");
    for(i = 0; i < n; i++)
        scanf("%d", &pages[i]);

    for(i = 0; i < m; i++)
```

```
{
    frames[i] = -1;
    time[i] = 0;
}

for(i = 0; i < n; i++)
{
    found = 0;

    for(j = 0; j < m; j++)
    {
        if(frames[j] == pages[i])
        {
            found = 1;
            time[j] = ++count;
            break;
        }
    }

    if(found == 0)
    {
        faults++;

        pos = -1;

        for(j = 0; j < m; j++)
        {
            if(frames[j] == -1)
            {
                pos = j;
                break;
            }
        }

        if(pos == -1)
        {
            pos = 0;
            min = time[0];

            for(j = 1; j < m; j++)
            {
                if(time[j] < min)
                {
                    min = time[j];
                    pos = j;
                }
            }
        }

        frames[pos] = pages[i];
        time[pos] = ++count;
```

```
        }

        printf("\n");
        for(j = 0; j < m; j++)
            printf("%d\t", frames[j]);
    }

    printf("\n\nTotal Page Faults = %d\n", faults);

    return 0;
}
```

OUTPUT:



RESULT:

Thus, the Least Recently Used (LRU) page replacement algorithm was implemented successfully using C in Fedora Linux, and the total number of page faults was calculated.